Conceptual Space Expansion Prompting

Experimental Process Documentation

A developmental investigation into how structured prompt parsing affects recitation-over-reasoning in large language models. Documents the iterative path from a bachelor thesis on Micro Interventions toward a paper-ready experimental framework, including full comparisons against three established CoT baselines (Standard, Self-Consistency, Tree of Thoughts).

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Subject: Conceptual Space Expansion Prompting (CSEP)

Target model: Claude Haiku 4.5

Judge model: Claude Opus 4.5

Contents

1. Executive Summary

This document records the complete experimental process conducted across a single extended session investigating a novel prompting technique called **Conceptual Space Expansion Prompting (CSEP)**. The work extends the author's bachelor thesis on Micro Interventions toward a journal-paper-level experimental framework.

The central experimental question emerged partway through: *Can a structured prompt-parsing pipeline prevent the recitation-over-reasoning failure mode that causes language models to produce canonical answers to subtly altered classical puzzles?*

Final Results (on 4 altered-classic puzzles)

Method	Passed	%	Primary mechanism
Baseline (direct)	0/4	0%	Pure recitation, no reasoning scaffolding
Self-Consistency (5 samples)	0/4	0%	Diversity sampling cannot escape systematic recitation
Standard CoT	1/4	25%	Step-by-step helps sometimes
CSEP v1 (hypothesis-first)	1/4	25%	Early hypothesis amplified recitation
Tree of Thoughts (real)	2/4	50%	Branching helps but evaluator shares training bias
Advanced CoT (hybrid)	2/4	50%	Diagnoses traps but cannot escape them
CSEP v2 (prompt-first)	4/4	100%	Concepts parsed before any answer

The Core Finding

Recitation-over-reasoning is not a failure of reasoning effort. Every established technique in the prompt engineering literature — Standard CoT, Self-Consistency, Tree of Thoughts, and advanced hybrid CoT — catches at most half of the traps. The decisive fix is **temporal separation of prompt parsing from answer generation**. When the model is forced to parse what the prompt explicitly says, analyze what it asks, and perform a pattern-match self-check — all before generating any tentative answer — the recitation trap is broken.

The most striking auxiliary finding is that Self-Consistency — widely cited as state-of-the-art for reasoning reliability — scores **worse than Standard CoT** on recitation problems. With temperature 0.7 and 5 independent samples, all 5 samples still recite the same canonical answer. Diversity sampling cannot escape a recitation failure mode that is systematic rather than stochastic.

2. Origin: The Bachelor Thesis

The work began with a review of the author's bachelor thesis *"Enhancing Large Language Models Output Using Micro Interventions"* (IU International University of Applied Sciences, March 2025). The thesis proposed that LLM outputs can be improved by prompting the model with small targeted probes that activate latent conceptual space before the main prompt is answered.

Theoretical Foundation

The thesis argues that three independent theoretical traditions converge on a single picture of how knowledge is organized and accessed:

Associativity	Psychoanalysis	Elements form associative complexes with mutual attraction/repulsion.
Felt Sense	Focusing	Holistic bodily awareness of an entire connected field around a concept.
Reference Frames	Thousand Brains Theory	Cortical columns build parallel, distributed representations of the same concept.

These converge on the concept of **Latent Conceptual Space (LCS)** — a richly connected region of knowledge that a prompt activates but that remains largely invisible in the final output. The thesis hypothesizes that LLMs operate on the same principle: a prompt induces an LCS from the weights, and only a small fraction of that space surfaces as tokens.

Micro Interventions (MI)

A Micro Intervention is the smallest usable probe of an LCS: a single additional prompt that asks the model to reflect on some aspect of the original question before answering. The thesis tested three MI types across five batches on ChatGPT 3.5 Turbo and reported 55% relative improvement in correct answers across batches 2–4.

Motivating Case: Kepler-438b

A particularly illuminating result involved a prompt asking the model to describe cultural practices of aliens on Kepler-438b. The model initially produced fabricated descriptions. When asked directly *"Would you answer this as fiction or as real?"*, it responded:

> *"I would answer the prompt as fiction because there is currently no scientific evidence or verified information about the existence of alien species on Kepler-438b or any other exoplanet."*

The model was not hallucinating in the sense of believing false things — it was producing confabulated answers despite internal awareness of their fictional status. A motivational problem rather than a cognitive one, foreshadowing the recitation phenomenon studied later in this document.

3. From MI to CSEP: The Conceptual Evolution

The session opened as a review of the thesis and gradually revealed a larger framework of which Micro Interventions were only the smallest unit.

Key Observations During the Discussion

1. Difference from Chain-of-Thought. CoT is procedural — it tells the model *how* to proceed. MI is conceptual — it tries to activate *what* the model already knows. These operate at different levels.

2. The diffusion model analogy. Generative image models produce hands with six fingers not because they don't know what a hand is, but because they attempt to produce an entire image in a single pass without decomposing the concept. Each component is well-known individually; the failure is combinatorial. LLMs exhibit the same pattern.

3. Zoom-in/zoom-out iteration. Human cognition alternates between holistic and granular views. Parallels Hawkins's cortical column voting and Gendlin's felt sense. The model should be able to do the same.

4. Imaginative bridging. When a concept is insufficiently defined, the model could construct a provisional idealized solution — explicitly flagged as imaginary — as a target vector for further reasoning.

5. Motivation and limbic gating. Human cognition requires motivational payoff to override default laziness. LLMs may have analogous structure in their training incentives — RLHF creates a "helpfulness gradient" that can push against truthful uncertainty.

6. Associative bridging. In creative problem-solving, the mind distills a problem to its minimal core and searches for analogues in unrelated domains. This expands the effective conceptual space around the problem.

7. Emotions as compression. Emotions are extreme compression of rich contextual information into scalar signals. The bachelor thesis contained a self-driving car analogy where a "fear index" trained on incident data modulated driving behavior. This could provide a pre-reasoning gate: "how dangerous is this prompt?" before deciding how to parse it.

The CSEP Hypothesis

From this discussion a specific testable method emerged: **Conceptual Space Expansion Prompting (CSEP)**. The central hypothesis:

> *Structured decomposition of a problem into its conceptual elements before generating an answer significantly improves LLM output quality compared to direct prompting and Chain-of-Thought.*

Scope Decision

The session explicitly bracketed several ambitious ideas for future work — zoom/unzoom iteration, imaginative bridging, motivational gating, analogical search across domains, and emotional compression signals — in favor of testing the minimal viable CSEP pipeline first.

4. Initial CSEP Pipeline Design

The first concrete pipeline, later retroactively labeled **CSEP v1**, had five phases. Each phase was a separate API call with its own prompt template.

Phase	Name	Task
1	Rough Hypothesis	Generate a tentative first-pass attempt at the answer
2	Master Concept	Identify the global meaning and category
3	Recursive Decomposition	Break into sub-concepts, up to three nested levels
4	Reintegration	Generate the answer using all prior analysis
5	Polishing	Review the answer against the decomposition

Key Design Decisions (v1)

Several design choices in v1 turned out later to be critical errors — errors that only became visible after empirical testing:

Hypothesis placed first. The rationale was that an early tentative hypothesis would give the model "something to anchor to" during decomposition. This was the core mistake. The rough hypothesis activated template-matching early, and everything that followed was rationalization.

No pattern-match check. The pipeline did not ask the model whether the current problem resembled a known canonical problem. There was no mechanism for the model to notice that it was reciting.

5. The Dataset Problem: Four Iterations

Before any meaningful test of CSEP was possible, a suitable dataset of failure cases was needed. This turned out to be far harder than anticipated.

Iteration 1: Ten Hand-Written Prompts

Problem: Haiku passed nearly all of them. Too easy.

Iteration 2: LLM-Generated 100-Question Set

Problem 1: reference answer contamination. The generator was asked to produce both the question and its correct answer in a single call. Opus consistently produced incorrect reference answers for math problems, contradicting itself within the same JSON object.

Problem 2: mode collapse. Opus called with the same prompt 100 times converged on the same surface template. Out of 20 math problems, 18 began with "Maria, who once won a blue ribbon at the county fair…". Out of 20 misleading-claim questions, 20 were variations on the 10%-of-brains myth.

Iteration 3: Two-Step Generation with Independent Solver

Restructured: Opus generates only question text, separate call solves it, third call re-verifies. Solved contamination. **Problem:** Mode collapse persisted.

Iteration 4: Hand-Curated Diverse Set

Claude was asked to hand-craft 30 diverse test questions with 30 distinct subtypes. **Result:** Haiku passed all 30. Haiku 4.5 is simply strong on standard distractor and false-premise problems.

Pivot: Documented Failure Modes

Strategy changed from "generate tricky questions" to "find problems the literature has already documented as LLM failure modes." A web search surfaced four highly relevant papers:

Paper	Key contribution
RoR-Bench (Yan et al., 2025)	"Recitation over Reasoning": subtly altering one word in classic problems causes 50–60% perfor
Easy Problems That LLMs Get Wrong (Williams & Huckle, 2024)	Curated modified classics where LLMs consistently recite the original.
Altered Riddles Dataset (marcodsn, 2025)	HuggingFace dataset of classic riddles with key details changed.
BrainBench (Tang, 2025)	Rental-car-return test case and related commonsense scenarios where Opus 4.6 achieves only 80

6. The Hard Dataset: Altered Classics

A 20-question dataset of "altered classics" was hand-crafted. Each question takes a well-known puzzle and changes one key condition so that the canonical answer no longer applies. Full dataset in *data/hard_questions.json*.

Four Examples That Became the Core Test Set

#1 Surgeon, father explicit:

> *"A man and his son are in a car accident. … The surgeon looks at the boy and says, 'I can operate on this boy — he is my son!'* ***The surgeon is the boy's father.*** *How is this possible?"*

The classic answer ("surgeon is the mother") is explicitly ruled out.

#5 Bat and ball, direct price:

> *"A bat and a ball cost $1.10 in total.* ***The bat costs $1.00.*** *How much does the ball cost?"*

The classic CRT says "$1.00 more than the ball" — answer $0.05. Here the bat's price is direct — answer $0.10.

#7 River crossing, no capacity limit:

> *"A man has a wolf, a goat, and a cabbage.* ***There is no boat. The river is shallow enough to wade through.*** *…"*

Classic solution requires the boat's one-item capacity. No boat = no capacity = carry all three at once.

#20 Locked room, all causes excluded:

> *"A man is found dead…* ***The coroner rules the death was NOT drowning, NOT glass injury, NOT poisoning, NOT heart attack.****"*

Classic answer is hanging from melted ice. Altered version rules out every common cause. Answer: insufficient information.

7. Baseline Testing on Altered Classics

The full 20-question dataset was run against Haiku 4.5 with a plain prompt. Tool: *haiku_hard_test.jsx*. Result: **16/20 correct, 4/20 failures**. The 4 failures — #1, #5, #7, #20 — became the core test set.

The Four Failures

#1 Surgeon:

> *"The surgeon is the boy's mother. The riddle assumes the surgeon must be male, but there's no reason that has to be the case."*

The model ignored the sentence "The surgeon is the boy's father" that literally preceded the question.

#5 Bat and ball:

> *"The ball costs $0.05 (5 cents). The bat costs $1.00 more than the ball… (The intuitive answer of $0.10 is a common error.)"*

The model fabricated the phrase "$1.00 more than the ball" and dismissed the correct answer as a "common error."

#7 No-boat river:

> *"Since there's no boat and the river is shallow enough to wade through, the man can simply carry each item across one at a time… The solution is the same sequence of moves, just done on foot."*

Read the alteration, verbalized it, and then dismissed it.

#20 Locked room:

> *"The most common intended answer is that the man stood on a block of ice to hang himself…"*

The prompt explicitly excluded every common cause. The model recited regardless.

8. CSEP v1: Does Decomposition Help?

The four failures were run through the CSEP v1 pipeline. Tool: *csep_vs_baseline.jsx*. Data: *data/csep_vs_baseline_4.json*.

Question	Baseline	CSEP v1	Fell for trap
#1 Surgeon	✗	✗	Yes
#5 Bat and ball	✗	✓	No
#7 No-boat river	✗	✗	Yes
#20 Locked room	✗	✗	Yes
Total	**0/4**	**1/4**	**3/4**

The Smoking Gun: Q#7 Decomposition

In phase 3 (decomposition), the model wrote:

> *"1.1 Transport Mechanism — The man wades through a shallow river, carrying one item at a time. This replaces the boat in the classic puzzle* ***but leaves the single-item-per-trip constraint entirely intact****, so the surface variation changes nothing about the logical structure."*

The model invented the phrase "one item at a time" and declared the single-item constraint "intact" — a constraint nowhere in the prompt. The decomposition structure gave this hallucination formal labeling (1.1, Level 1) and therefore authority.

Dialectical Collapse: Q#1 Surgeon

Phase 1 generated *two* hypotheses: "same-sex couple" and "surgeon is mother." Phase 2 promoted "mother" to master concept. Phase 3 labeled "two fathers" as secondary. Phase 4 finalized "mother." Phase 5 confirmed. The model arrived at the correct answer in phase 1, then spent four phases talking itself out of it.

9. The Negative Result That Changed Everything

The CSEP v1 result (1/4) was a negative finding that turned out to be the most valuable moment of the entire session:

> *Structured decomposition, performed after a tentative hypothesis has been generated, does not mitigate recitation-over-reasoning. In some cases it amplifies it.*

Why CSEP v1 Fails

1. The phase-1 hypothesis commits the model to a pattern. When asked for a hypothesis before parsing the prompt carefully, the model template-matches to the nearest known problem. Once this happens, the rest of the pipeline operates inside the frame the template provides.

2. Subsequent phases rationalize rather than critique. Master-concept generalizes the hypothesized frame. Decomposition enumerates sub-aspects *of the frame*, not of the prompt. Reintegration writes the answer consistent with the frame. Polishing checks the answer against the decomposition — but the decomposition inherited the frame's assumptions, so the check is circular.

3. Formal structure gives the reciting answer epistemic weight. A plain baseline "the surgeon is the mother" is dismissable. The same answer wrapped in five phases of hierarchical analysis looks *more* correct, not less.

Design Principles for CSEP v2

Principle 1 — No hypothesis before parsing. First phase must extract concepts from the prompt text, not generate any answer attempt.

Principle 2 — Explicit pattern-match self-check. Dedicated phase asking whether the problem resembles a canonical version, and if so, comparing condition-by-condition.

Principle 3 — Final verification against explicit concepts only. Last phase must flag any condition, fact, or assumption that is not in the prompt.

10. CSEP v2: Prompt-First Pipeline

CSEP v2 redesigned the five phases from first principles. Tool: *csep_v2_test.jsx*. Data: *data/csep_v2_results.json*.

Phase	Name	Task
1	Concepts in Prompt	List explicit and implicit concepts, mark source
2	What is Asked	Classify type, specify answer shape, list load-bearing conditions
3	Pattern-Match Self-Check	Compare to canonical version; identify differences
4	Answer	Generate answer, constrained to prompt-only information
5	Verification	Check for imported conditions and contradictions

Phase 3 (highlighted) is the critical addition.

Results: 4/4

Question	Baseline	CSEP v1	CSEP v2
#1 Surgeon	✗	✗	✓
#5 Bat and ball	✗	✓	✓
#7 No-boat river	✗	✗	✓
#20 Locked room	✗	✗	✓
Total	**0/4**	**1/4**	**4/4**

Phase 3 Produced Crisp Self-Diagnosis

Phase 3 outputs for all four questions:

#5 (bat and ball):

> *"The canonical version says the bat costs **$1.00 more than the ball** — a relative, comparative statement. The current problem says the bat costs **$1.00** — an absolute, fixed price. … The irony is that $0.10 — the answer everyone mocks as the 'dumb intuitive trap' in the canonical puzzle — is actually the correct answer to this version."*

#7 (river crossing):

> *"The one-at-a-time carrying constraint, which is the mechanical heart of the canonical puzzle, has been neither stated nor replaced. It simply isn't there."*

#20 (locked room):

> *"If the answer is 'he stood on a block of ice and hanged himself,' the water is explained by melting ice — but the broken glass is entirely unaccounted for. … The canonical answer simply ignores it."*

#1 (surgeon):

> *"The problem instead explicitly states as given fact that 'the surgeon is the boy's father' — which means the puzzle is no longer asking the reader to discover who the surgeon is. It has already told you."*

In all four cases, phase 3 produced the diagnosis needed to avoid the trap. This is the mechanism CSEP v2 buys: a structural requirement that the model check for recitation *before* it commits to any answer.

11. The Three CoT Baselines: Full SOTA Comparison

To make the CSEP v2 claim defensible, it was tested against three established prompting techniques from the literature. Tool: *three_cot_baselines.jsx*. Data: *data/three_cot_baselines.json*.

The Three Baselines

Method	Source	Mechanism
Standard CoT	Wei et al. 2022	1 API call, "Let's think step by step"
Self-Consistency	Wang et al. 2022	5 samples at temperature 0.7, majority vote
Tree of Thoughts	Yao et al. 2023	3 branches, independent expansion, evaluator picks best

Full Results

Question	Base	Std CoT	Self-Cons	ToT	CSEP v1	CSEP v2
#1 Surgeon	✗	✗	✗	✗	✗	✓
#5 Bat and ball	✗	✗	✗	✓	✓	✓
#7 No-boat river	✗	✓	✗	✓	✗	✓
#20 Locked room	✗	✗	✗	✗	✗	✓
Total	**0/4**	**1/4**	**0/4**	**2/4**	**1/4**	**4/4**
%	**0%**	**25%**	**0%**	**50%**	**25%**	**100%**

Full SOTA comparison across six methods on four altered-classic puzzles.

The Self-Consistency Paradox

The most surprising result is that **Self-Consistency scored 0/4 — worse than Standard CoT**. Self-Consistency (Wang et al. 2022) is one of the most-cited reasoning-reliability techniques, routinely cited as SOTA for mathematical reasoning. Yet on recitation problems, it fails completely.

The mechanism is visible in the vote counts. For #5 (bat and ball):

> ***All 5 samples:*** *"The ball costs $0.05 (5 cents)."*
> ***Vote count:*** *{"$0.05": 5}*
> ***Majority: $0.05 — recitation wins 5-0.***

Even at temperature 0.7, all five samples produce identical wrong answers. The model's attachment to the canonical template is so strong that diversity sampling does not shake it. Similarly for #1 surgeon: all 5 samples say "the surgeon is the boy's mother" despite the prompt explicitly stating the surgeon is the father.

This is evidence that recitation is systematic rather than stochastic. Stochastic errors would give varied answers across samples; a majority vote would recover the correct one. Systematic template-matching produces identical errors, and majority vote amplifies rather than corrects them.

The Q#7 Degradation Case

A particularly revealing case is Q#7 (no-boat river). **Standard CoT passed**: the single CoT run produced a careful reading that correctly noticed the "no boat" condition and concluded the man can wade across with all three. But **Self-Consistency with 5 samples failed**. The vote distribution:

> ***4 samples:*** *classic multi-trip sequence (goat, wolf, bring goat back, cabbage, goat)*
> ***1 sample:*** *wade across with all three at once*
> ***Majority: classic multi-trip — wrong by 4 to 1.***

One sample did find the correct answer, but it was outvoted 4-1 by samples that recited the classic. **Self-Consistency can degrade a single-run success into a majority failure** when the recitation template is the dominant attractor in the sampling distribution.

The Tree of Thoughts Evaluator Bias

Tree of Thoughts performed better (2/4), but its failures revealed a subtle problem. For Q#1 surgeon, ToT evaluator gave Branch B score 9/10 with this justification:

> *"Branch B most cleanly identifies and dismantles the single hidden assumption driving the paradox — that 'surgeon' implies male — and delivers the classic intended answer (the surgeon is the boy's mother) without unnecessary tangents"*

The evaluator itself fell for the recitation. It praised the classic answer as "the cleanest analysis" despite the prompt explicitly contradicting it. When evaluator and solver share the same training bias, ToT's multi-path architecture provides no protection. Both components happily arrive at the same trap.

For Q#5 bat-and-ball, ToT did work — the evaluator picked Branch B which correctly identified the altered phrasing. This succeeded by luck: the trick-lens branch happened to produce the right answer, and the evaluator happened to score it highest. On Q#1 the same architecture produced exactly the opposite outcome.

Summary of the Three-Baseline Comparison

The three established baselines reveal that recitation-over-reasoning resists every orthodox approach:

Standard CoT (1/4): Sometimes the step-by-step approach notices the alteration; sometimes it does not. No systematic mechanism.

Self-Consistency (0/4): Diversity sampling amplifies the dominant template. Can actively degrade single-run successes.

Tree of Thoughts (2/4): Multi-branch exploration helps when the branch selector happens to score correctly, but both generator and evaluator share training biases that cause them to score the recited pattern highest.

None of these techniques has a mechanism that forces the model to parse the prompt explicitly before generating tentative answers. CSEP v2 does, and achieves 4/4.

12. Final Results & Interpretation

Consolidated Results Table

Method	API calls	Result	Why
Baseline	1	0/4	Pure recitation
Standard CoT	1	1/4	Step-by-step occasionally catches alteration
Self-Consistency	6	0/4	Recitation is systematic, not stochastic — majority vote amplifies it
CSEP v1	5	1/4	Hypothesis-first commits to template early
Tree of Thoughts	6	2/4	Branching helps but evaluator shares training bias
Advanced CoT	1	2/4	Diagnoses traps but single-pass cannot escape
CSEP v2	5	4/4	Prompt parsed before any answer is generated

Interpretation

The pattern across these methods suggests that recitation-over-reasoning has a specific structure:

1. It is not fixable by more reasoning alone. Advanced CoT with adversarial self-check catches only half the traps. Tree of Thoughts with multiple independent branches does no better.

2. It is not fixable by diversity sampling. Self-Consistency actively underperforms Standard CoT. Temperature-based sampling cannot reach answers that lie outside the template's attractor basin.

3. It is not fixable by post-hoc critique. Both Advanced CoT's adversarial check and CSEP v1's polishing phase demonstrate that a critical review applied after a tentative answer is insufficient. The critique itself inherits the biased frame.

4. It is fixable by pre-reasoning prompt parsing. CSEP v2 achieves the fix by requiring the model to parse the prompt, meta-analyze the question, and check for pattern-match risk *before* generating any answer.

The Working Theory

Recitation fires at the moment the model begins generating an answer. If that moment arrives with an unparsed prompt and a recognized template, the template wins. No amount of subsequent reasoning, branching, voting, or critiquing can reliably dislodge an already-committed template — because each of those operations inherits the template's frame.

If that moment arrives with an already-analyzed prompt (concepts explicitly listed) and an already-diagnosed pattern-match risk (phase 3), the template's activation is weakened. The model can reason against the template rather than from it. This is temporal separation of parsing from generation.

Implications for Future Work

Scaling the experiment. Immediate next step: run all seven methods on the full 20-question altered-classics dataset. Current 4-question subset is proof-of-mechanism, not a performance benchmark.

Over-correction control. CSEP v2 might cause the model to treat all questions as potential trick questions, degrading performance on straightforward ones. A control set is needed.

Ablation study. CSEP v2 has three new features (prompt-first ordering, meta-analysis, pattern-match check). Ablating each would identify which is doing the work.

The parked ideas return. Zoom/unzoom iteration, imaginative bridging, motivational gating ("fear index"), analogical search, and emotional compression signals now fit into a coherent research program built on the CSEP v2 finding.

Limitations

Four questions is not a statistical sample. One target model (Haiku 4.5) and one judge model (Opus 4.5) do not generalize across the model landscape. The 20-question altered-classics set itself needs independent validation. None of these limitations changes the core finding — the diagnostic contrast between the baselines and CSEP v2 is *within-question* and reproducible, not a statistical artifact.

13. Appendix A: Tool Inventory

All tools are React JSX files intended to run in the Claude Artifacts environment. Each connects to the Anthropic API via browser fetch.

File	Purpose
`csep_pipeline_test.jsx`	Single-question CSEP v1 test
`csep_batch_test.jsx`	Early 10-question batch (no judging)
`csep_batch_evaluated.jsx`	Adds Opus judge to batch test
`question_filter.jsx`	LLM-generated questions, iteration 2
`question_filter_v2.jsx`	Two-step generation with independent solver
`haiku_curated_test.jsx`	Tests Haiku on 30 hand-curated questions
`haiku_hard_test.jsx`	Baseline on 20 altered classics
`haiku_baseline_100.jsx`	Early 100-question baseline
`csep_vs_baseline.jsx`	CSEP v1 on the 4 failures
`csep_v2_test.jsx`	CSEP v2 prompt-first pipeline
`cot_test.jsx`	Standard CoT tool (included separately)
`advanced_cot_test.jsx`	Hybrid ToT + adversarial CoT
`three_cot_baselines.jsx`	Standard + Self-Consistency + real ToT, all three baselines

14. Appendix B: Data Inventory

File	Contents
`hard_questions.json`	20 altered classics (core test set)
`curated_questions.json`	30 hand-crafted diverse questions
`questions.json`	100 LLM-generated questions (iteration 2)
`all_questions_60.json`	60 Opus-generated questions showing mode collapse
`csep_test_questions.json`	5 failures from iteration 2
`csep_test_questions2_.json`	3 failures from iteration 3
`haiku_hard_failures.json`	4 Haiku baseline failures (the core test set)
`csep_vs_baseline_4.json`	CSEP v1 results (1/4 fixed)
`csep_v2_results.json`	CSEP v2 results (4/4 fixed)
`advanced_cot_results.json`	Advanced hybrid CoT results (2/4 fixed)
`three_cot_baselines.json`	Standard CoT + Self-Consistency + ToT results

Core Dataset Files for Paper

hard_questions.json — the 20 altered-classics test set.
csep_v2_results.json — phase-3 self-diagnosis quotes for paper.
three_cot_baselines.json — full SOTA comparison data.
advanced_cot_results.json — advanced CoT reasoning traces.

15. Appendix C: Reproducibility Guide

Environment

All experiments ran in-browser against the Anthropic Messages API. No local Python required — only an Anthropic API key and a browser that can execute JSX as Claude Artifacts.

Models

Target: `claude-haiku-4-5-20251001`
Judge / Generator: `claude-opus-4-5-20250929`

Step-by-Step Reproduction

1. Verify baseline failures. Run *haiku_hard_test.jsx* on 20 altered classics. Expected: ~16/20 correct, 4/20 failures (IDs #1, #5, #7, #20).

2. Run CSEP v1. Open *csep_vs_baseline.jsx*. Expected: 1/4 success (#5 only).

3. Run CSEP v2. Open *csep_v2_test.jsx*. Expected: 4/4 success.

4. Run three CoT baselines. Open *three_cot_baselines.jsx*. Expected: Standard 1/4, Self-Consistency 0/4, Tree of Thoughts 2/4.

5. Run advanced hybrid CoT. Open *advanced_cot_test.jsx*. Expected: 2/4.

6. Inspect phase 3 outputs of CSEP v2. Each question's phase 3 should show clear self-diagnosis of recitation risk.

Variation

Variation between runs is expected due to temperature and documented non-determinism. For formal statistical study, each condition should run at least 10 times. The current results are proof-of-mechanism; qualitative claim (baselines fail systematically, CSEP v2 succeeds systematically) should be robust to run-to-run variation.

Extensions

• Scale to all 20 altered classics (currently 4-question subset).
• Control set of normal questions for over-correction check.
• Ablation: prompt-first ordering vs meta-analysis vs pattern-match check.
• Different target models (Sonnet, Opus, GPT-4, Gemini) for generalization.
• Parked ideas: zoom/unzoom, imaginative bridging, motivational gating, analogical search — each as follow-up experiments.

End of documentation.